

Algebraic Identities Worksheet

Q1: $(3n+5)^2$
 $= (3n)^2 + (5)^2 + 2(3n)(5)$
 $= \boxed{9n^2 + 25 + 30n}$
 $[(a+b)^2 = a^2 + b^2 + 2ab]$

Q2: $(4a-7b)^2$
 $= (4a)^2 + (7b)^2 - 2(4a)(7b)$
 $= \boxed{16a^2 + 49b^2 - 56ab}$
 $[(a-b)^2 = a^2 + b^2 - 2ab]$

Q3: 97×103
 $= (100-3)(100+3)$
 $= 100^2 - 3^2$
 $= 10000 - 9$
 $= \boxed{9991}$
 $[(a-b)(a+b) = a^2 - b^2]$

Q4: $(y+9)(y-4)$
 $= y^2 + y(9-4) + (9)(-4)$
 $= \boxed{y^2 + 5y - 36}$

Q5: $(2p+3)^3$
 $= (2p)^3 + \cancel{3(2p)^2(3)} + 3(2p)^2(3) + 3(2p)(3)^2 + (3)^3$
 $= \boxed{8p^3 + \cancel{12p^2} + 36p^2 + 54p + 27}$

Q6: $a+b = 9, ab = 20$
 $(a+b)^2 = (9)^2$
 $\Rightarrow a^2 + b^2 + 2ab = 81$
 $\Rightarrow a^2 + b^2 + 2(20) = 81$
 $\Rightarrow \boxed{a^2 + b^2 = 41}$

Q7: $a - b = 4$, $ab = 2$

$$(a - b)^2 = 4^2$$

$$\Rightarrow a^2 + b^2 - 2ab = 16$$

$$\Rightarrow a^2 + b^2 - 2(2) = 16$$

$$\Rightarrow \boxed{a^2 + b^2 = 58}$$

Q8: ~~$a + b = 8$, $ab = 15$~~

~~$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$
$$= (8)^2$$~~

Q8: $a + b = 8$, $ab = 15$

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$\Rightarrow (8)^2 = a^2 + b^2 + 2(15)$$

$$\Rightarrow a^2 + b^2 = 64 - 30$$

$$\Rightarrow a^2 + b^2 = 34$$

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$\Rightarrow a^3 + b^3 = (8)(34 - 15)$$

$$\Rightarrow a^3 + b^3 = (8)(19)$$

$$\Rightarrow \boxed{a^3 + b^3 = 152}$$

Q9: $a - b = 5$, $ab = 14$

$$(a - b)^2 = a^2 + b^2 - 2ab$$

$$\Rightarrow (5)^2 = a^2 + b^2 - 2(14)$$

$$\Rightarrow a^2 + b^2 = 25 + 28$$

$$\Rightarrow a^2 + b^2 = 53$$

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$$

$$\Rightarrow a^3 - b^3 = (5)(53 + 14)$$

$$\Rightarrow a^3 - b^3 = 5(67)$$

$$\Rightarrow \boxed{a^3 - b^3 = 335}$$

$$\begin{array}{r} 2556 \\ 14 \\ \hline 2224 \\ 556 \times \\ \hline 7784 \end{array}$$

$$\begin{array}{r} 370 \\ 37 \\ \hline 407 \end{array}$$

Q10: $15^3 + 5^3 = (15+5)[(15)^2 + (5)^2 - 15(5)]$
 $= (20)(225 + 25 - 75)$
 $= 20(175)$

$$\begin{aligned} a^3 + b^3 \\ = (a+b)(a^2 + b^2 - ab) \end{aligned}$$

$\Rightarrow \boxed{15^3 + 5^3 = 3500}$

Q11: $20^3 - 6^3 = (20-6)[(20)^2 + (6)^2 + 20(6)]$
 $\Rightarrow 20^3 - 6^3 = (14)(400 + 36 + 120)$
 $\Rightarrow 20^3 - 6^3 = (14)(556)$
 $\Rightarrow \boxed{20^3 - 6^3 = 7784}$

$$\begin{aligned} a^3 - b^3 \\ = (a-b)(a^2 + b^2 - ab) \end{aligned}$$

Q12: $a+b+c=11$, $ab+bc+ca=20$
 $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
 $\Rightarrow (11)^2 = a^2 + b^2 + c^2 + 2(20)$
 $\Rightarrow a^2 + b^2 + c^2 = 121 - 40$
 $\Rightarrow \boxed{a^2 + b^2 + c^2 = 81}$

Q13: $a+b=11$, $a^2+b^2=65$, $a^3+b^3=?$
 $(a+b)^2 = a^2 + b^2 + 2ab$
 $\Rightarrow (11)^2 = 65 + 2ab$
 $\Rightarrow ab = \frac{121-65}{2} = \frac{56}{2} = 28$

$a^3+b^3 = (a+b)(a^2+b^2-ab)$
 $\Rightarrow a^3+b^3 = (11)(65-28)$
 $\Rightarrow a^3+b^3 = (11)(37)$
 $\Rightarrow \boxed{a^3+b^3 = 407}$

~~Q14: $a-b=6$, $a^2+b^2=68$, $a^3-b^3=?$~~

$$Q14: a-b=6, a^2+b^2=68$$

$$a^3-b^3=?$$

$$(a-b)^2 = a^2 + b^2 - 2ab$$

$$\Rightarrow (6)^2 = 68 - 2(ab)$$

$$\Rightarrow ab = \frac{36 - 68}{-2} = 16$$

$$a^3-b^3 = (a-b)(a^2+b^2+ab)$$

$$\Rightarrow a^3-b^3 = (6)(68+16)$$

$$\Rightarrow a^3-b^3 = (6)(84)$$

$$\Rightarrow \boxed{a^3-b^3 = 504}$$

$$Q15: a+b+c=13, ab+bc+ca=52$$

$$abc=60, a^3+b^3+c^3=?$$

$$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$$

$$\Rightarrow (13)^2 = a^2+b^2+c^2+2(52)$$

$$\Rightarrow a^2+b^2+c^2 = 169 - 104$$

$$\Rightarrow a^2+b^2+c^2 = 65$$

$$a^3+b^3+c^3-3abc = (a+b+c)$$

$$(a^2+b^2+c^2-ab-bc-ca)$$

$$\Rightarrow a^3+b^3+c^3-3(60) = (13)(65-52)$$

$$\Rightarrow a^3+b^3+c^3 = (13)(13) + (180)$$

$$\Rightarrow \boxed{a^3+b^3+c^3 = 349}$$